Fields and Waves I

Lecture 12

Electric Boundary Conditions & Capacitance

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Materials from other sources are referenced where they are used. Those listed as Ulaby are figures from Ulaby's textbook.

Overview

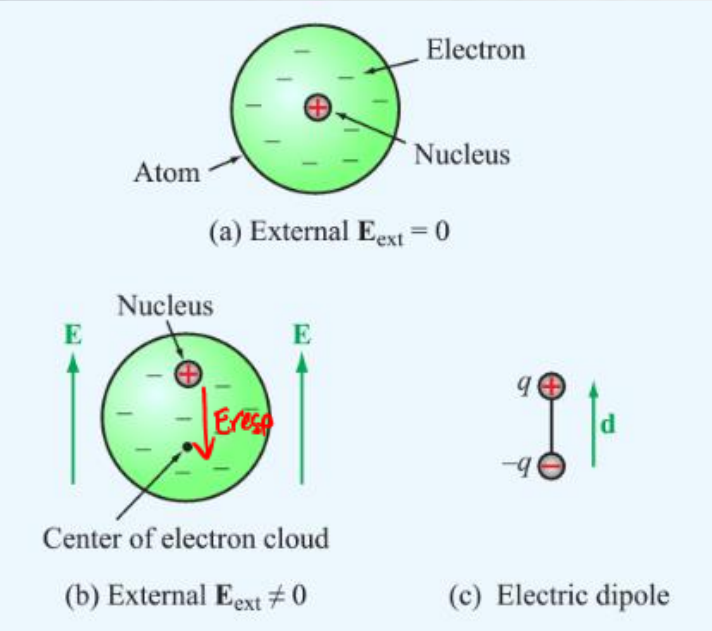
- Electric Boundary Conditions
- Capacitance

Review: Electric Materials

- In free space: $\mathbf{D} = \epsilon_0 \mathbf{E}$ $\epsilon = \epsilon_0$
- In a dielectric: $\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}$ $\epsilon = \epsilon_r \epsilon_0$

$\mathbf{E}$ applied

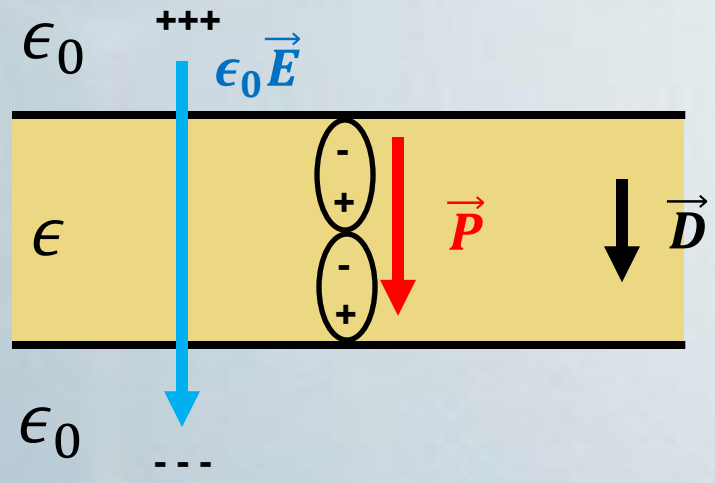
$\mathbf{P} = \epsilon_0 \chi \mathbf{E} = \rho_b \mathbf{d}$



Ulaby

Review: Electric Materials

Consider a parallel plate capacitor with a material of electric permittivity ϵ inside and ϵ_0 outside.



1. We apply an electric field $\vec{E}$ across the plates.

2. Polarization causes the formation of a *polarization density* field $\vec{P}$ that opposes $\vec{E}$

3. The electric displacement field $\vec{D}$ arises due to the remaining net charge density in the dielectric

$$\epsilon_0\vec{E} + \vec{P} = \vec{D}$$

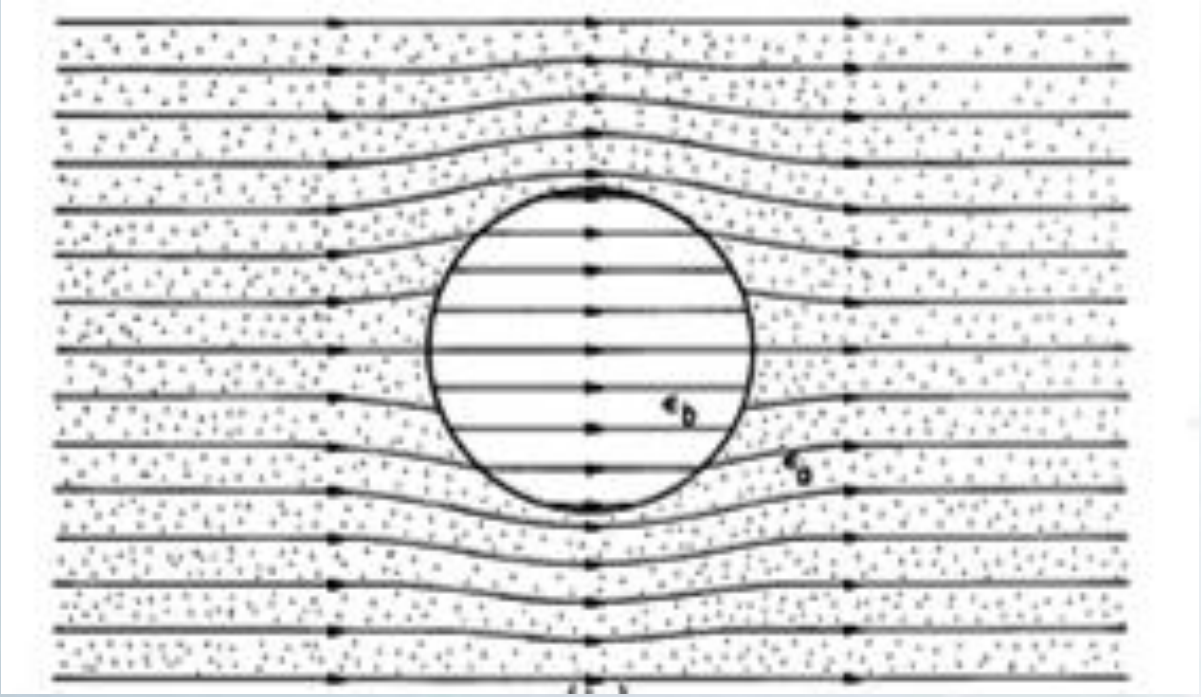
$$\rho_f + \rho_b = \epsilon_0\vec{\nabla}\cdot\vec{E}$$

$$-\vec{\nabla}\cdot\vec{P}$$

$$\vec{\nabla}\cdot\left(\epsilon_0\vec{E} + \vec{P}\right) = \vec{\nabla}\cdot\vec{D} = \rho_f$$

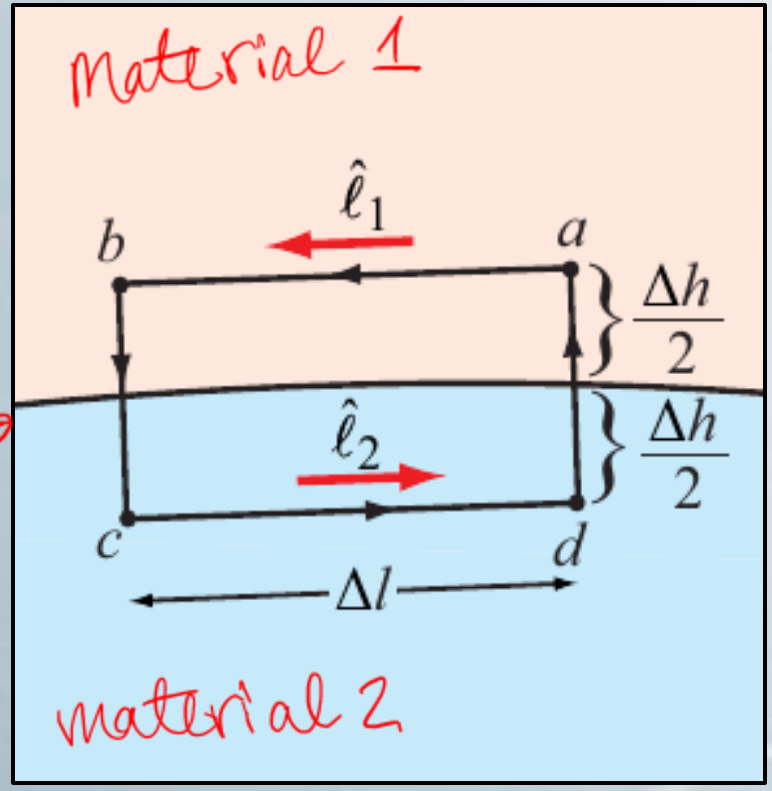
Electric Boundary Conditions

Boundary Conditions



Look at this picture again. How do electric fields behave at the boundary between two different dielectrics?

Boundary Conditions



interface

Ulaby

- We know that

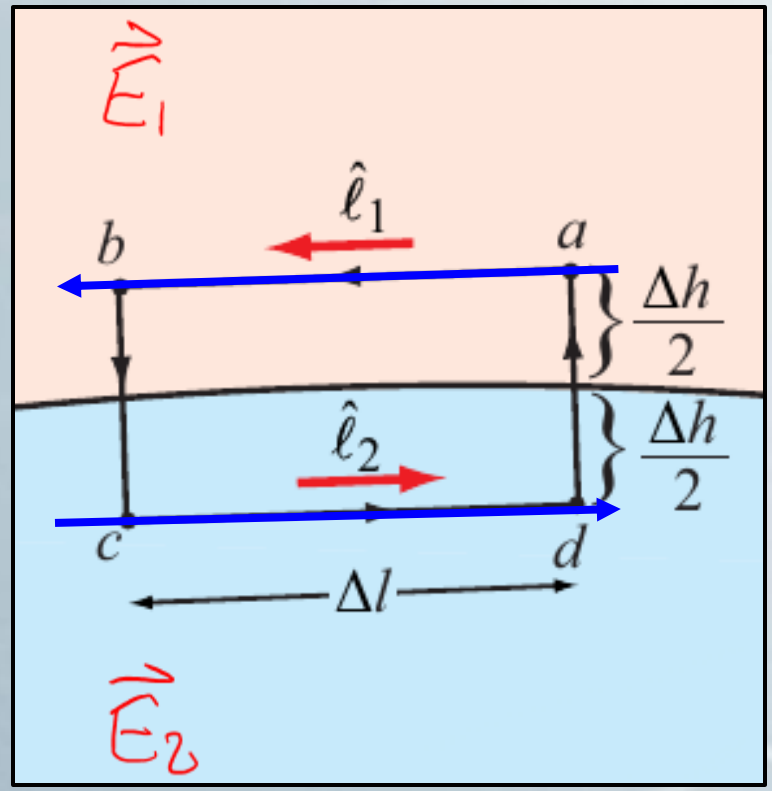
$$\oint \vec{E} \cdot \vec{dl} = 0$$

$\nabla \times \vec{E} \approx 0$

Curl of E = 0

- This will hold for any loop we choose, so we can choose $\Delta h \rightarrow 0$ so that the contributions of segments **bc** and **da** go to zero.
- Note that we have chosen $\hat{l}_1$ and $\hat{l}_2$ tangent to the surface.

Boundary Conditions

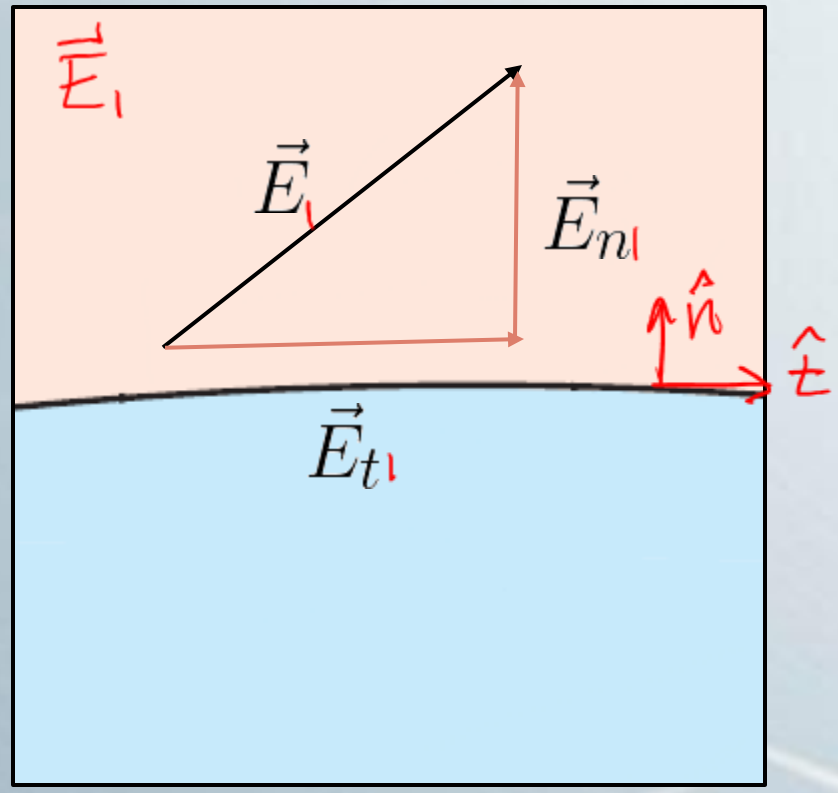


Ulaby

- We can then write

$$\int_a^b \vec{E}_1 \cdot \mathrm{d}\hat{l}_1 + \int_c^d \vec{E}_2 \cdot \mathrm{d}\hat{l}_2 = 0$$

Boundary Conditions



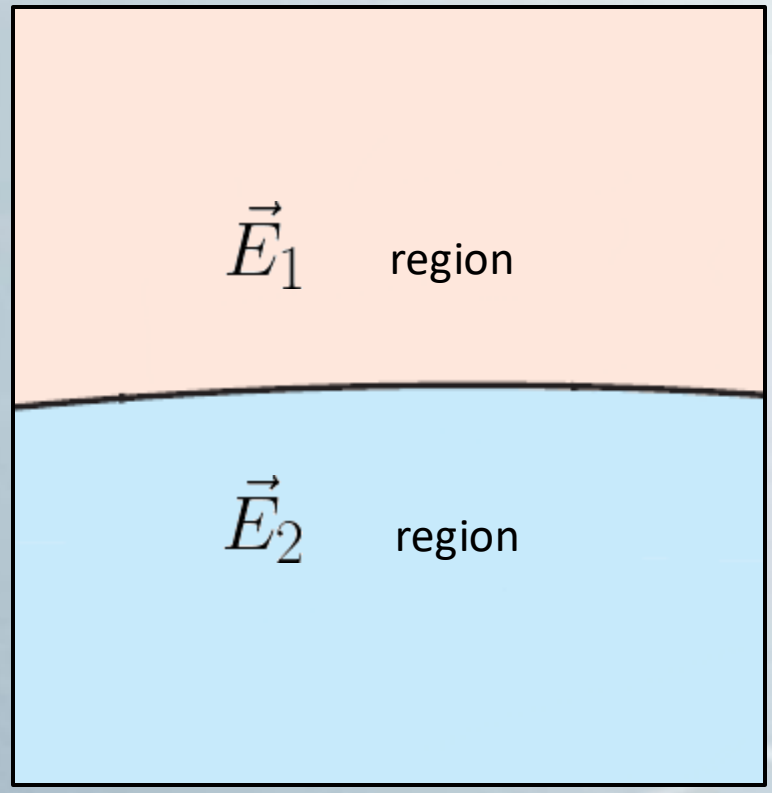
- Note that we can easily decompose any E-field vector into its tangent and normal components relative to the material boundary

$$\vec{E}_1 = \vec{E}_{n1} + \vec{E}_{t1}$$

$\vec{E}_{n1} = (\vec{E}_1 \cdot \hat{n})\hat{n}$ (normal)

$\vec{E}_{t1} = (\vec{E}_1 \cdot \hat{t})\hat{t}$ (tangent)

Boundary Conditions



- Therefore we can write

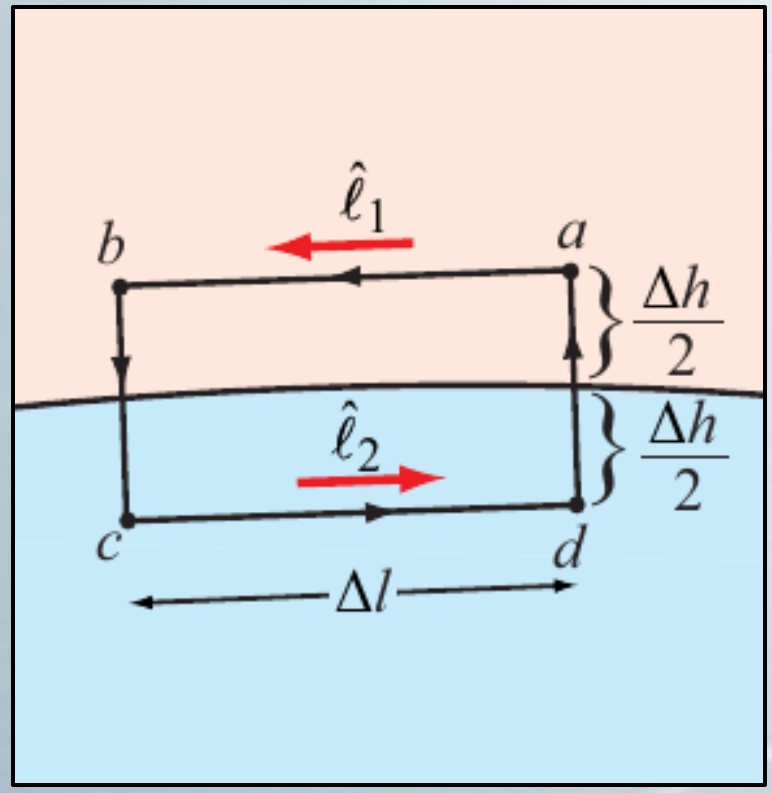
$$\vec{E}_1 = \vec{E}_{1t} + \vec{E}_{1n}$$

$$\vec{E}_2 = \vec{E}_{2t} + \vec{E}_{2n}$$

Boundary Conditions

Tangential Component

Tangential Component



Ulaby

- This equation:

$$\int_a^b \vec{E}_1 \cdot \mathrm{d}\hat{l}_1 + \int_c^d \vec{E}_2 \cdot \mathrm{d}\hat{l}_2 = 0$$

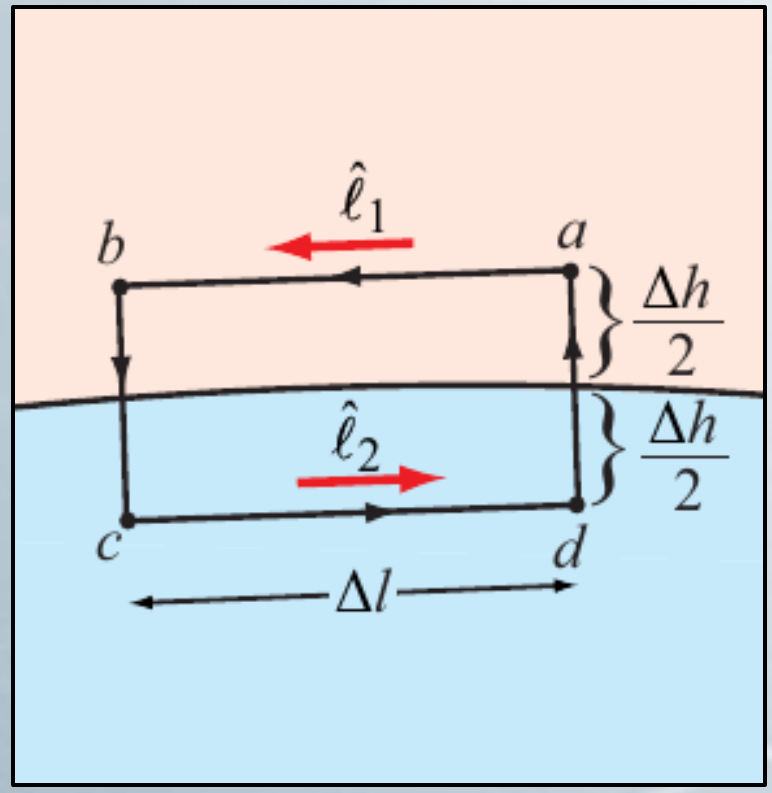
- then becomes:

$$(\vec{E}_1 - \vec{E}_2) \cdot \hat{l}_1 = 0$$

because

$$\hat{l}_1 = -\hat{l}_2$$

Boundary Conditions



Ulaby

- We chose $\hat{l}_1$ and $\hat{l}_2$ such that

$$\vec{E}_1 \cdot \hat{l}_1 = \vec{E}_{1t}$$

$$\vec{E}_2 \cdot \hat{l}_2 = \vec{E}_{2t}$$

- Now we simplify:

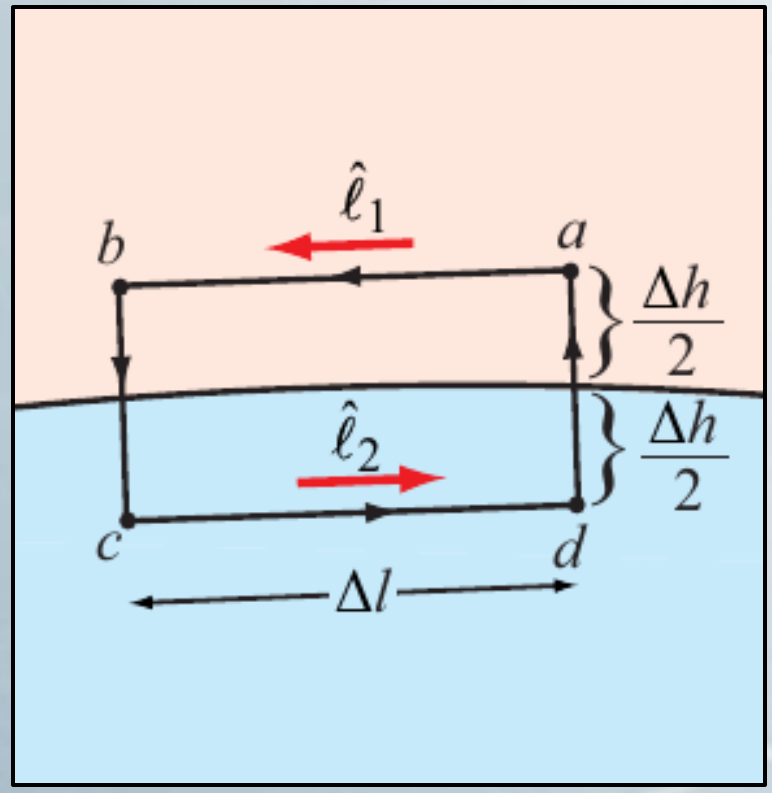
from integral

$$(\vec{E}_1 - \vec{E}_2) \cdot \hat{l}_1 = 0$$

$$\vec{E}_{1t} = \vec{E}_{2t}$$

Boundary Conditions

Tangential Component



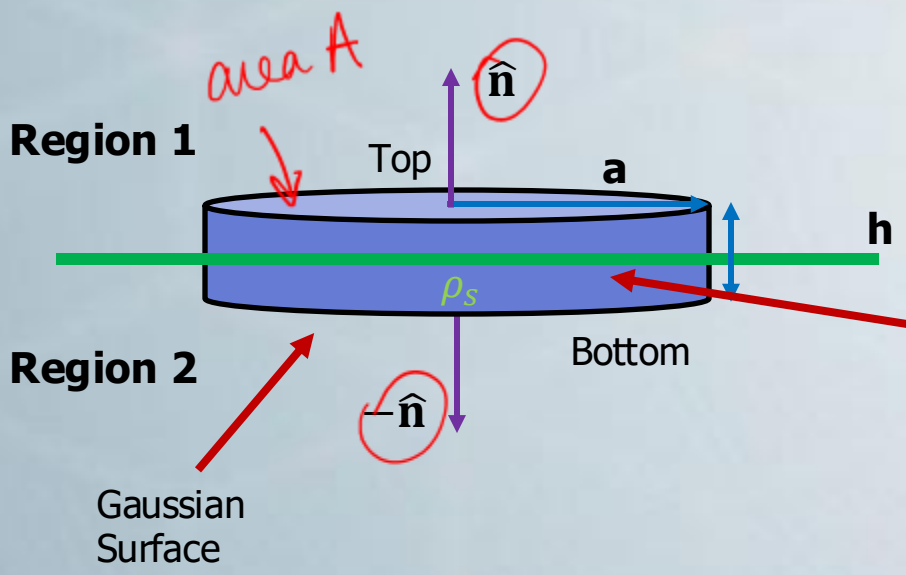
Ulaby

$$\vec{E}_{1t} = \vec{E}_{2t}$$

- So the **component of the E-field that is tangent** to a media boundary **is continuous across it**.
- What about normal to the boundary?

Boundary Conditions

Normal Component



- We will use Gauss's law to determine the normal boundary condition

$$\oint \mathbf{D} \cdot d\mathbf{S} = Q_{enclosed}$$

- At the interface between the two materials, there is in general a surface charge density ρ_s

Assuming $h \ll a$ (thin disk)

$$Q_{enclosed} = \rho_s A$$

- Performing the flux integrals of Gauss's law:

$$\oint \mathbf{D} \cdot d\mathbf{S} = \oint \mathbf{D}_1 \cdot d\mathbf{S}_{\text{top}} + \oint \mathbf{D}_2 \cdot d\mathbf{S}_{\text{bot}}$$

$$= A[\mathbf{D}_1 \cdot \hat{\mathbf{n}}] + A[\mathbf{D}_2 \cdot (-\hat{\mathbf{n}})]$$

$$= A\mathrm{D}_{1\mathrm{n}} - A\mathrm{D}_{2\mathrm{n}} = \rho_s A$$

Therefore:

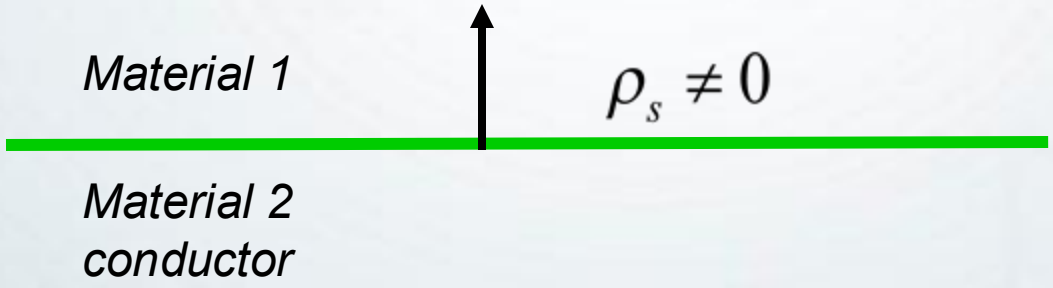
$$\boxed{\mathrm{D}_{1\mathrm{n}} - \mathrm{D}_{2\mathrm{n}} = \rho_s}$$

Boundary Conditions

Conductors vs. Dielectrics

Case 1: REGION 2 is a **CONDUCTOR**, $D_2 = E_2 = 0$

$$\therefore D_{1n} = \rho_s$$

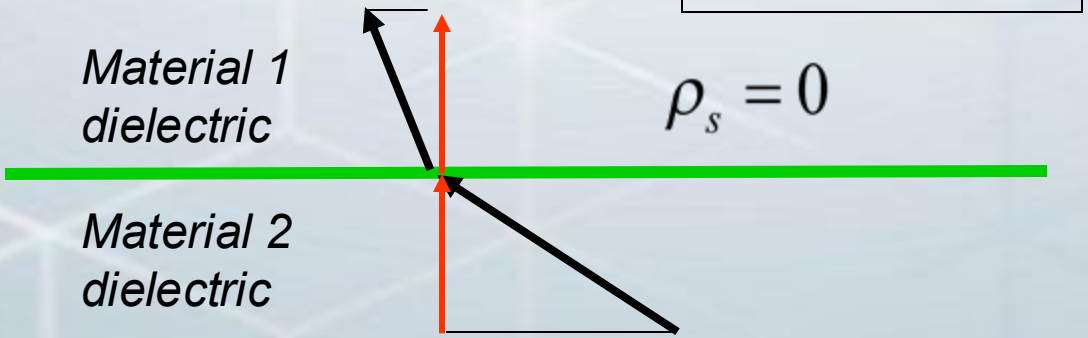


Can only really get ρ_s with conductors: free charge

Case 2: REGIONS 1 & 2 are **DIELECTRICS** with $\rho_s = 0$

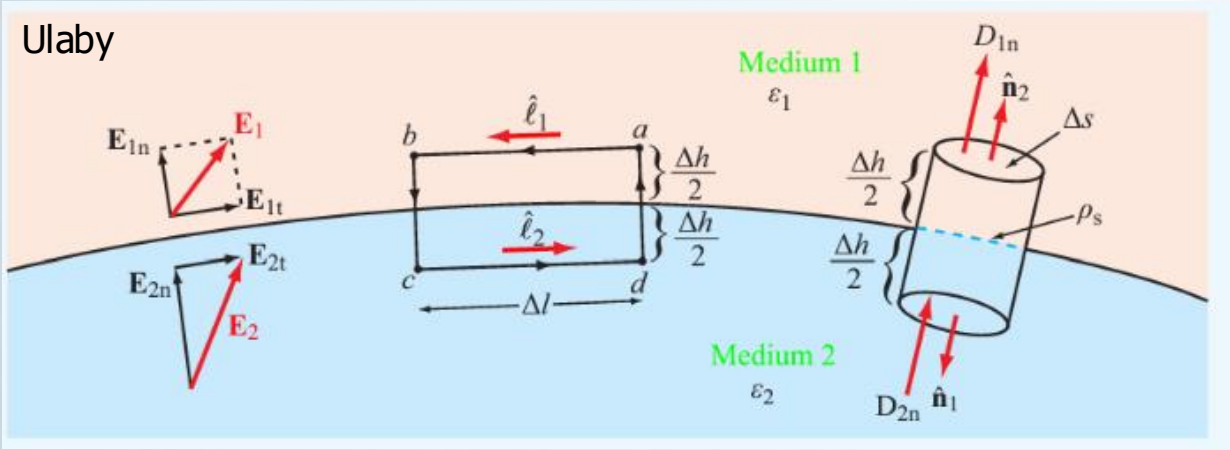
$$\therefore D_{1n} = D_{2n}$$

$$\therefore \varepsilon_1 E_{1n} = \varepsilon_2 E_{2n}$$



Boundary Conditions

Summary



$$\mathbf{E} = \mathrm{E_t}\hat{\mathbf{t}} + \mathrm{E_n}\hat{\mathbf{n}} \qquad \mathbf{D} = \mathrm{D_t}\hat{\mathbf{t}} + \mathrm{D_n}\hat{\mathbf{n}}$$

Tangential component: parallel to boundary

$$E_{1t} = E_{2t}$$

Tangential E field component must be continuous across boundary

Normal component: perpendicular to boundary

$$D_{1n} - D_{2n} = \rho_s$$

Normal D field component flux must be continuous across boundary

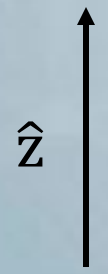
up to a difference due to ρ_s

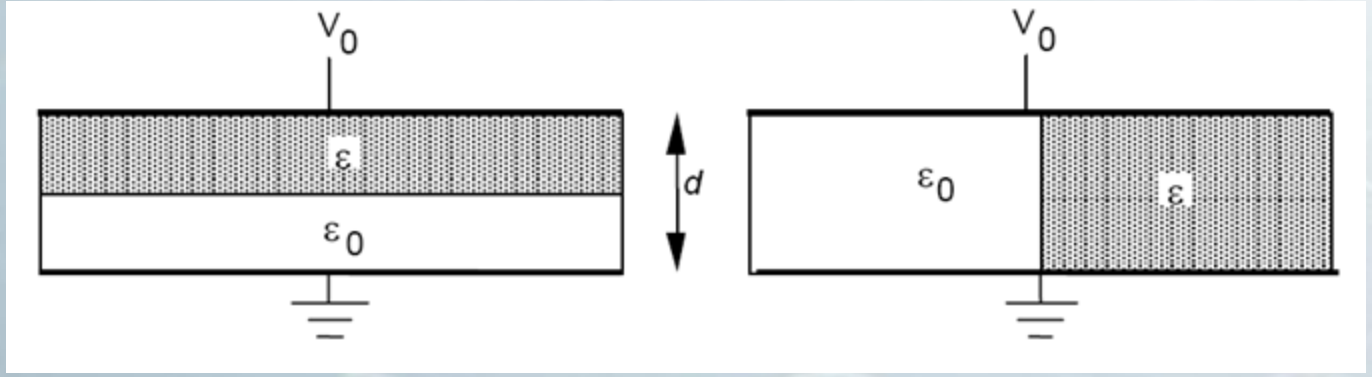
Boundary Conditions

Consider the two parallel plate geometries below. Assume that the plate dimensions are large compared to the separation *d* and ignore fringe effects. For the two figures, the electric field in the air region, (specified by ε_o) is given by:

$$\vec{E}_{air} = -\frac{V_0}{d}\left(\frac{2\epsilon_r}{1+\epsilon_r}\right)\hat{z}$$ figure on left

$$\vec{E}_{air} = -\frac{V_0}{d}\hat{z}$$ figure on right





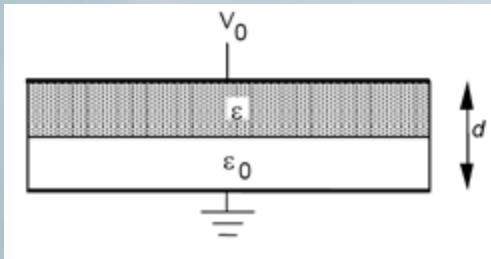
Boundary Conditions

1. For both cases, find **E** in the dielectric region. Find **D** in both regions. Within a given region, **D** and **E** do not vary with position.

2. Find the charge density on the plates at all locations.

Boundary Conditions

Case #1



$$\vec{E}_{\text{air}} = -\frac{V_0}{d}\left(\frac{2\epsilon_r}{1+\epsilon_r}\right)\hat{z}$$

Electrostatic Boundary Conditions

Case #1:



$\vec{E}_{air} = -\frac{V_0}{d}\left(\frac{2\epsilon_r}{1+\epsilon_r}\right)\hat{z}$

$E_{1t} = E_{2t}$

$D_{1n} - D_{2n} = \rho_s$ ← mat 1 is on top

- Find $\vec{E}_{dielectric}$, $\vec{D}_{air}$, and $\vec{D}_{dielectric}$
- Find $\vec{D}_{air} = \epsilon_0 \vec{E}_{air} = \boxed{-\frac{V_0}{d}\left(\frac{2\epsilon_r\epsilon_0}{1+\epsilon_r}\right)\hat{z}}$
- Decompose $\vec{E}_{air}$ and $\vec{D}_{air}$ into normal and tangential components

$\vec{E}_{air,t} = 0$; $\vec{E}_{air,n} = -\frac{V_0}{d}\left(\frac{2\epsilon_r}{1+\epsilon_r}\right)\hat{z}$

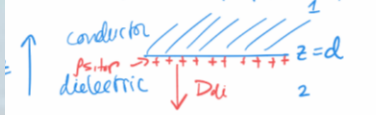
$\vec{D}_{air,t} = 0$; $\vec{D}_{air,n} = -\frac{V_0}{d}\left(\frac{2\epsilon_r\epsilon_0}{1+\epsilon_r}\right)\hat{z}$

- Tangential fields: $\vec{E}_{di,t} = \vec{E}_{air,t} = 0$
- Normal fields: $D_{di,n} - D_{air,n} = \rho_s^{\,0} \rightarrow D_{di,n} = D_{air,n} = -\frac{V_0}{d}\left(\frac{2\epsilon_r\epsilon_0}{1+\epsilon_r}\right)$
- Full fields in dielectric:

$$\vec{D}_{di} = -\frac{V_0}{d}\left(\frac{2\epsilon_r\epsilon_0}{1+\epsilon_r}\right)\hat{z}$$

$$\vec{E}_{di} = -\frac{V_0}{d}\left(\frac{2}{1+\epsilon_r}\right)\hat{z}$$

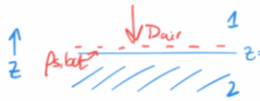
- Find ρ_s on both plates: use $D_{1n} - D_{2n} = \rho_s$ at the dielectric/conductor boundaries
- For the top plate ($\rho_{s,top}$):



$D_{conductor,n}^{\,0} - D_{di,n} = \rho_{s,top}$

$\rho_{s,top} = -D_{di,n}(d) = \frac{V_0}{d}\left(\frac{2\epsilon_r\epsilon_0}{1+\epsilon_r}\right)$

- For the bottom plate ($\rho_{s,bot}$):

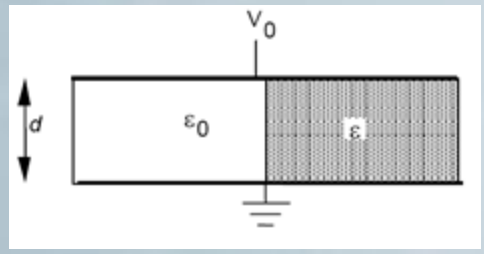


$D_{air,n} - D_{conductor,n}^{\,0} = \rho_{s,bot}$

$\rho_{s,bot} = -\frac{V_0}{d}\left(\frac{2\epsilon_r\epsilon_0}{1+\epsilon_r}\right)$

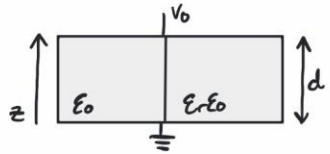
Boundary Conditions

Case #2



$$\vec{E}_{\text{air}} = -\frac{V_0}{d}\hat{z}$$

Case #2:



$\vec{E}_{air} = -\frac{V_0}{d}\hat{z}$

- Find $\vec{E}_{dielec}$, $\vec{D}_{air}$, and $\vec{D}_{dielec}$

$\vec{E}_{air,t} = -\frac{V_0}{d}\hat{z}$, so $\boxed{\vec{E}_{dielec,t} = -\frac{V_0}{d}\hat{z}}$ and $\boxed{\vec{D}_{dielec,z} = -\frac{\varepsilon_r\varepsilon_0 V_0}{d}\hat{z}}$

$\boxed{\vec{D}_{air} = -\frac{\varepsilon_0 V_0}{d}\hat{z}}$

- Find ρ_s on both plates
  - For the top plate: $D_{in}^{\,0} - D_{zn} = \rho_{s,top}$

    so $\rho_{s,top} = \begin{cases} \frac{\varepsilon_0 V_0}{d} & \text{above air region} \\ \frac{\varepsilon_r\varepsilon_0 V_0}{d} & \text{above dielectric region} \end{cases}$
  - For the bottom plate: $D_{in} - D_{zn}^{\,0} = \rho_{s,bot}$

    so $\rho_{s,bot} = \begin{cases} -\frac{\varepsilon_0 V_0}{d} & \text{below air region} \\ -\frac{\varepsilon_r\varepsilon_0 V_0}{d} & \text{below dielectric region} \end{cases}$

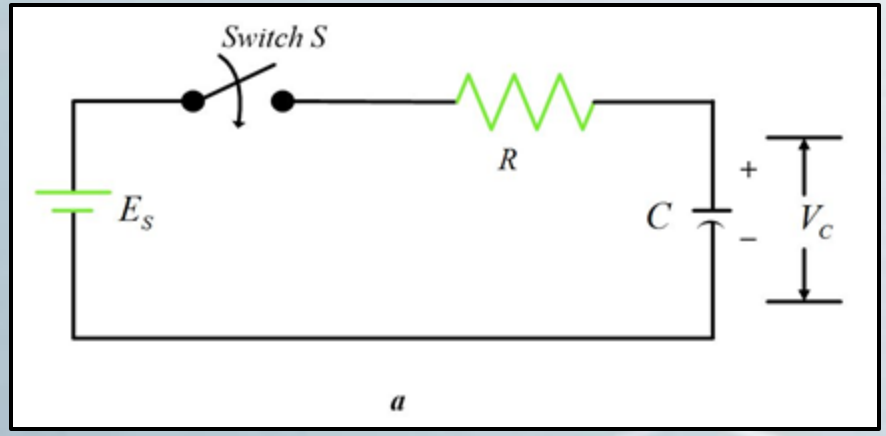
Boundary Conditions

Do Lecture 12 Exercise 1 in groups of up to 4.

Capacitance

Capacitance

- In earlier EE classes you did analysis on circuits containing capacitors, which have a property called capacitance. What is capacitance?





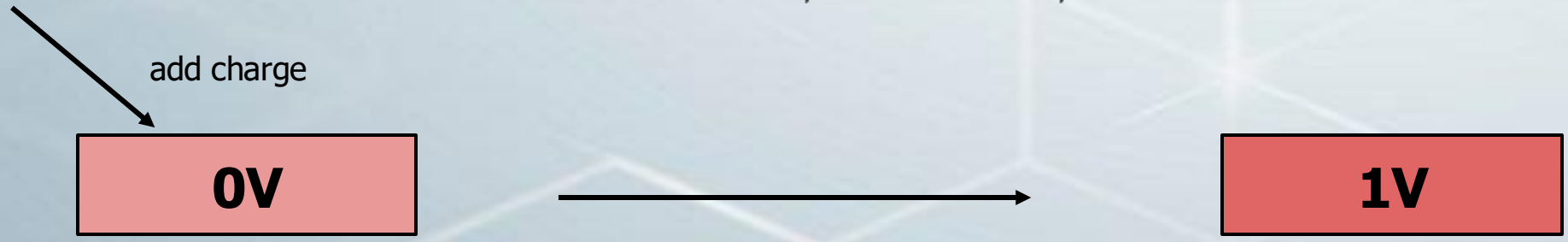
Capacitance

- Capacitance describes how much the stored charge in a system changes (**ΔQ**) when there is a change in potential difference in the system (**ΔV**)
- There are two types of capacitance: self-capacitance and mutual capacitance.

Capacitance

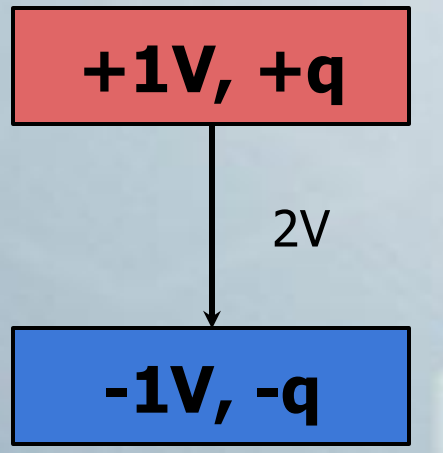
- **Self-capacitance** describes how much charge is required to increase the electric potential of some conductor by 1V
- In this case there is *no parallel plate*. We just need a reference point from which to define the voltage of the conductor.

$$C = \frac{q}{V} = \frac{dq/dt}{dV/dt} = \frac{i(t)}{dV/dt}$$



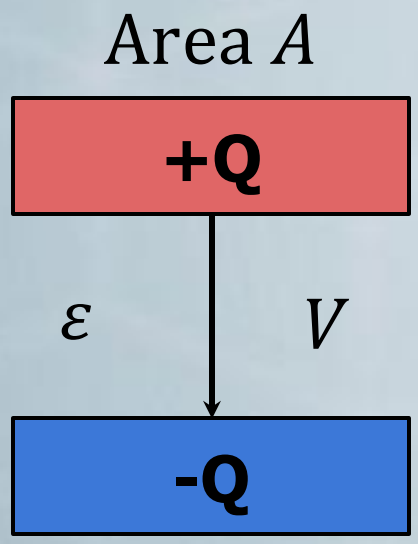
Capacitance

- **Mutual capacitance**: the charge-voltage relationship between two conductors. (This applies to the parallel plate capacitors that you are familiar with.)
- In this case, the charge in the capacitance equation is an equal and opposite charge on *both* plates, and the voltage is *between* them.



$$C = \frac{q}{V} = \frac{dq/dt}{dV/dt} = \frac{i(t)}{dV/dt}$$

Capacitance



$$\frac{Q}{V} = C = \varepsilon \cdot \frac{A}{d}$$

Parallel Plate Capacitance

To get large C,

- increase A
- increase ε
- decrease d

This is how electrolytics increase C

(differential capacitance $C' = \frac{dQ}{dV}$)

Define: $C = \frac{Q}{V}$ ← charge on 1 conductor

ΔV between conductors

Note that: $\frac{d}{dt}(Q = CV) \Rightarrow I = C\frac{dV}{dt}$

Capacitance

- The energy stored in capacitors is stored in the electric field
- The stored electrical energy is:

$$W_e = \frac{1}{2}CV^2$$

- If we substitute values of C and V for a parallel plate capacitor:

$$W_e = \frac{1}{2}CV^2 = \frac{1}{2}\cdot\left(\varepsilon\frac{A}{d}\right)\left(|\vec{E}|d\right)^2 = \frac{1}{2}\cdot\underbrace{\varepsilon|\vec{E}|^2}_{\text{Energy density}}\cdot\underbrace{Ad}_{\text{Volume}}$$

Capacitance

- In general, we can write the total stored energy as:

$$W_e = \frac{1}{2}\iiint \varepsilon \left|\vec{E}\right|^2 dV$$

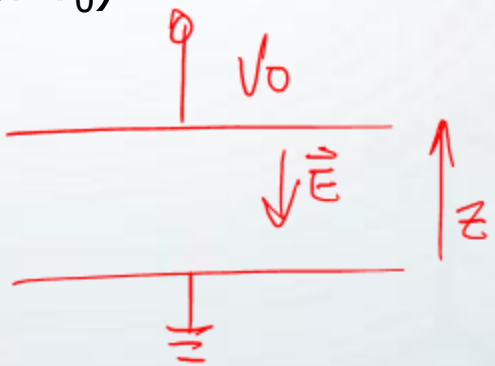
or

$$W_e = \frac{1}{2}\iiint \left(\vec{D}\cdot\vec{E}\right) dV$$

Capacitance

- Use the energy formulation to compute C for the parallel plate capacitor

We know that $\vec{E} = -\frac{V_0}{d}\hat{z}$ (we need $\vec{E}$ in terms of V_0)



- Compute total energy $W_e = \frac{1}{2}\int (\vec{D}\cdot\vec{E})dV$

$$W_e = \frac{1}{2}\int \left(-\epsilon\frac{V_0}{d}\hat{z}\right)\cdot\left(-\frac{V_0}{d}\hat{z}\right)\mathrm{d}x\mathrm{d}y\mathrm{d}z = \epsilon\frac{Ad}{2}\frac{V_0^2}{d^2}$$

$$\epsilon\frac{Ad}{2}\frac{V_0^2}{d^2} = \frac{1}{2}CV_0^2 \rightarrow \boldsymbol{C = \frac{\epsilon A}{d}}$$

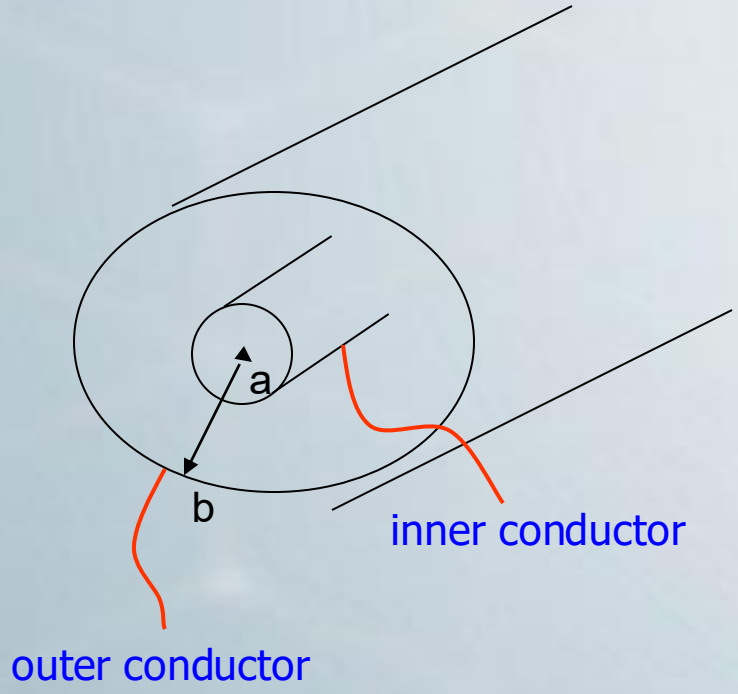
Capacitance

- Consider a coaxial cable with inner radius, *a*, and outer radius, *b*, length *l*.
- The outer conductor is grounded, and inner conductor has voltage $V = V_0$.
- The relation between surface charge density on the inner conductor and V_0 is

$$\rho_s = \frac{\epsilon V_0}{a \ln\left(\frac{b}{a}\right)}$$

1. Find the capacitance using $C = Q/V$ (or $C = dQ/dV$ for differential capacitance).
2. Find the stored energy in the system by integrating the energy density over the system volume.
3. Find the capacitance using the stored energy from part 2.

Capacitance



- For a coaxial cable:

$$V_{ab} = \frac{\rho_s a}{\epsilon} \ln\left(\frac{b}{a}\right)$$

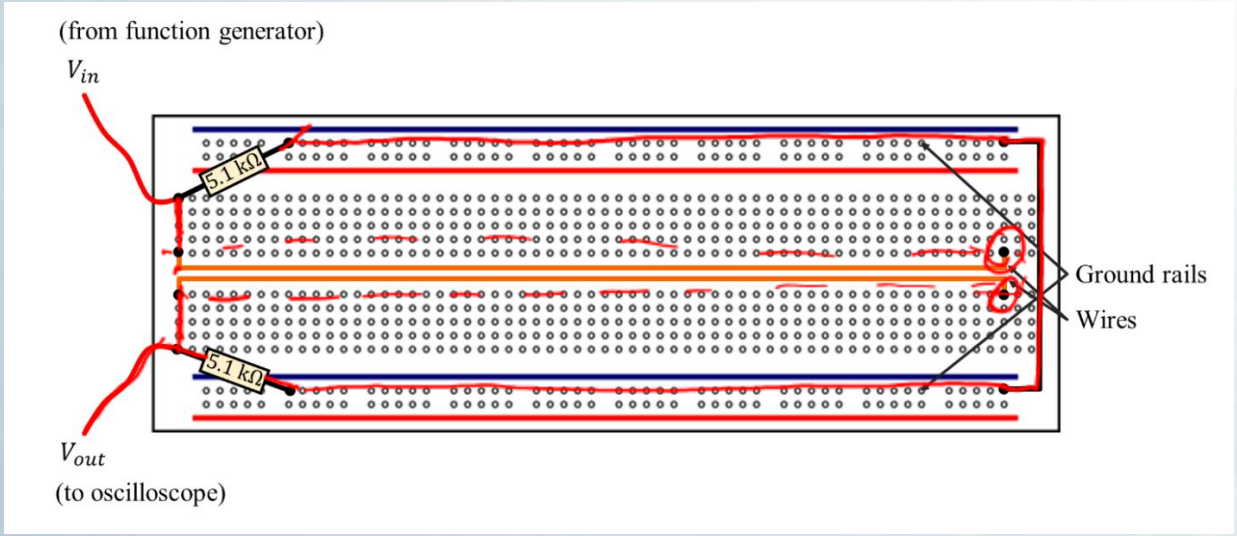
- The electric field is given by

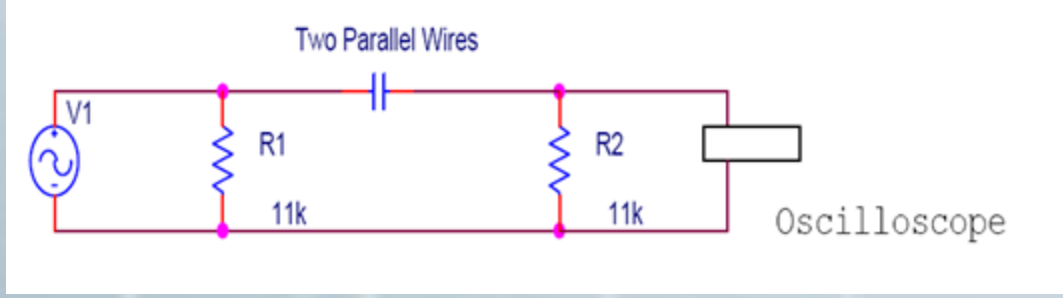
$$\vec{\mathbf{E}} = \frac{V_0}{r \ln\left(\frac{b}{a}\right)} \hat{\mathbf{r}}$$

Capacitance

Capacitance

- Any 2 conductors have capacitance (such as wires on a breadboard)
- How do we estimate the capacitance of a system like this?



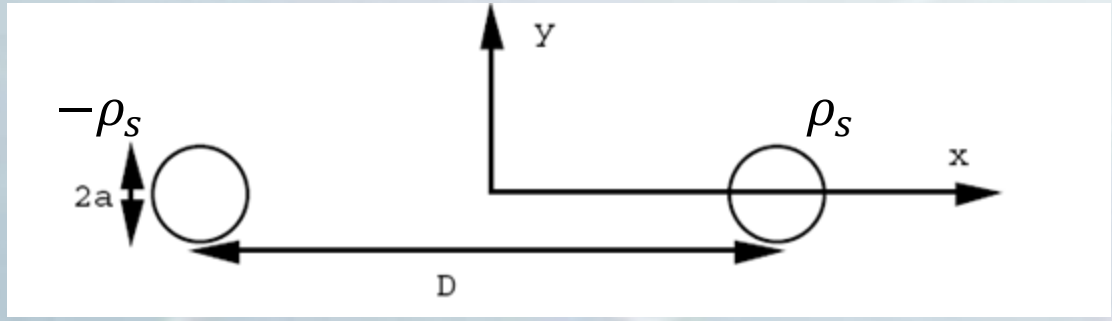


Capacitance

The objective of this problem is to determine the capacitance between 2 conducting wires.

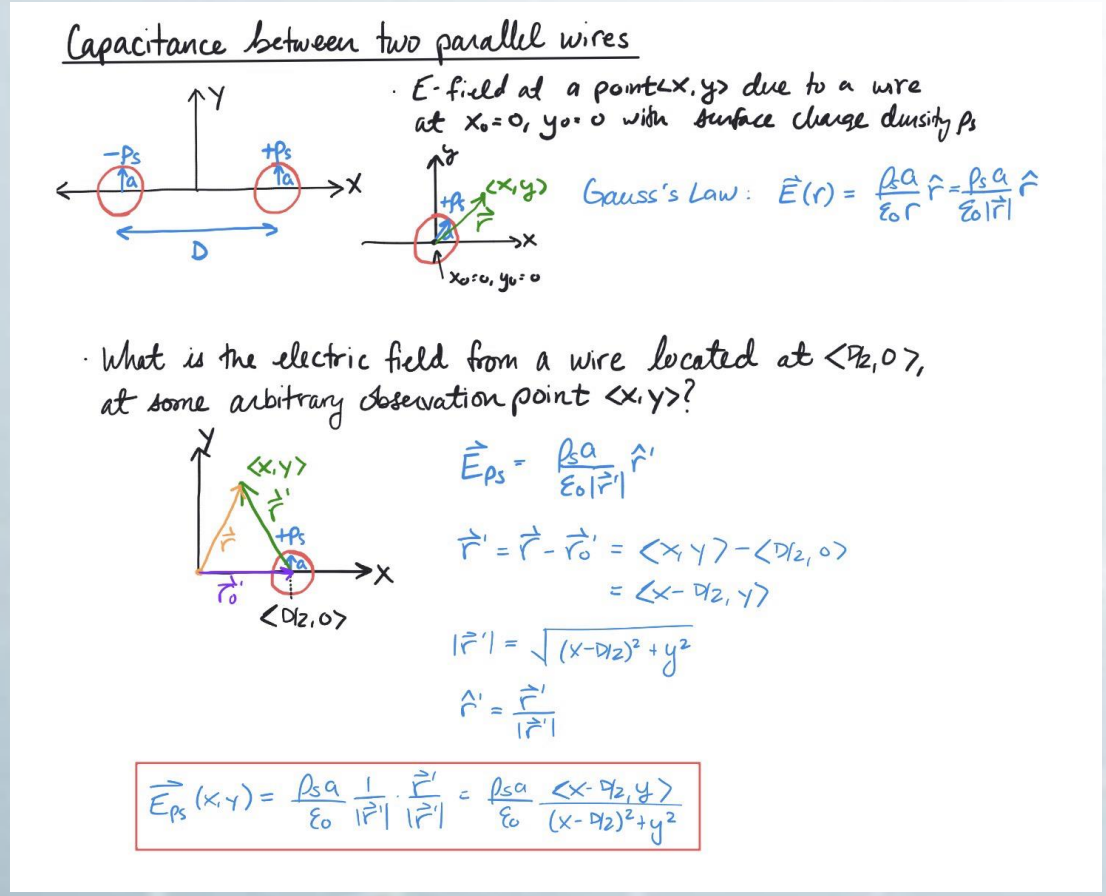
We will assume there are uniform surface charge densities of equal magnitude and opposite sign on the two wires.

- What is the potential difference between the wires?
- What is the capacitance ?



Capacitance

- Electric field at observation point $x = x$ and $y = y$ for a single wire located at $x = x_0$ and $y = 0$



Capacitance

- What is the electric field due to both wires?

$$\vec{E} = \vec{E}_{ps} + \vec{E}_{-ps}$$

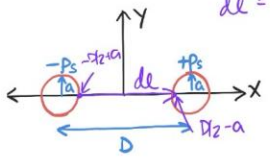
wire at $x = -D/2$: $\vec{E}_{ps}(x,y)$: at $\langle -D/2, 0\rangle$ and has $\rho = -\rho_s$ $(\vec{r}' = \langle x + D/2, y\rangle)$

$$\vec{E}_{ps} = -\frac{\rho_s a}{\varepsilon_0}\frac{\langle x+D/2, y\rangle}{(x+D/2)^2+y^2}$$

$$\boxed{\vec{E}(x,y) = \frac{\rho_s a}{\varepsilon_0}\left\{\frac{\langle x-D/2, y\rangle}{(x-D/2)^2+y^2} - \frac{\langle x+D/2, y\rangle}{(x+D/2)^2+y^2}\right\}}$$

- What is the potential difference between them?

$d\vec{l} = \hat{x}\,dx$ at $y = 0$



$$V(D/2-a) - V(-D/2+a) = -\int_{-D/2+a}^{D/2-a}\frac{\rho_s a}{\varepsilon_0}\left(\frac{\langle x-D/2, 0\rangle}{(x-D/2)^2+0^2} - \frac{\langle x+D/2, 0\rangle}{(x+D/2)^2+0^2}\right)\cdot\hat{x}\,dx$$

$$= \frac{\rho_s a}{\varepsilon_0}\int_{-D/2+a}^{D/2-a}\left(\frac{1}{x+D/2} - \frac{1}{x-D/2}\right)dx =$$

$$= \frac{\rho_s a}{\varepsilon_0}\left[\ln(x+D/2) - \ln(x-D/2)\right]\Big|_{-D/2+a}^{D/2-a}$$

$$= \frac{\rho_s a}{\varepsilon_0}\left[\ln\left(\frac{x+D/2}{x-D/2}\right)\right]\Big|_{-D/2+a}^{D/2-a} = \frac{\rho_s a}{\varepsilon_0}\left[\ln\left(\frac{D-a}{-a}\right) - \ln\left(\frac{a}{a-D}\right)\right]$$

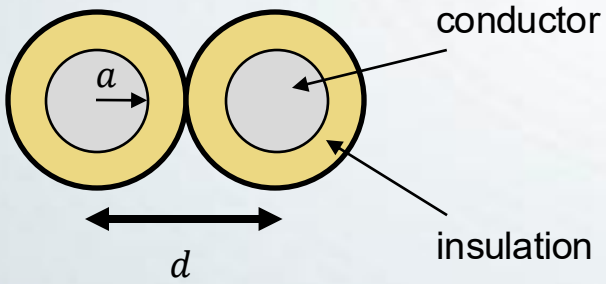
$$= \frac{\rho_s a}{\varepsilon_0}\left[\ln\left(\left(\frac{a-D}{a}\right)^2\right)\right] = \boxed{\frac{2\rho_s a}{\varepsilon_0}\ln\left(\left|\frac{a-D}{a}\right|\right)}$$

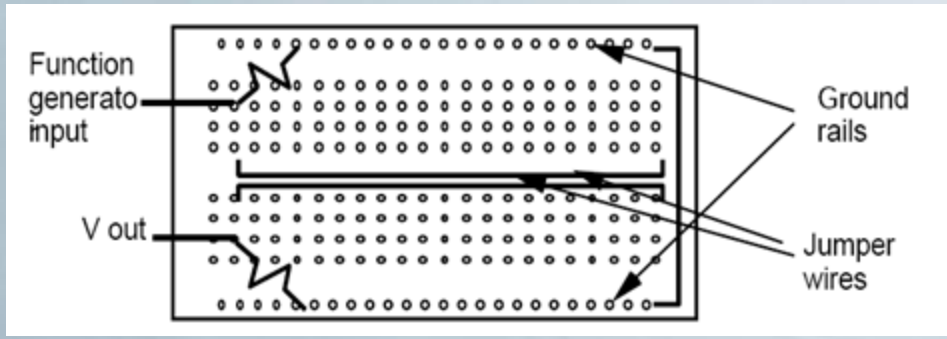
- What is the capacitance?

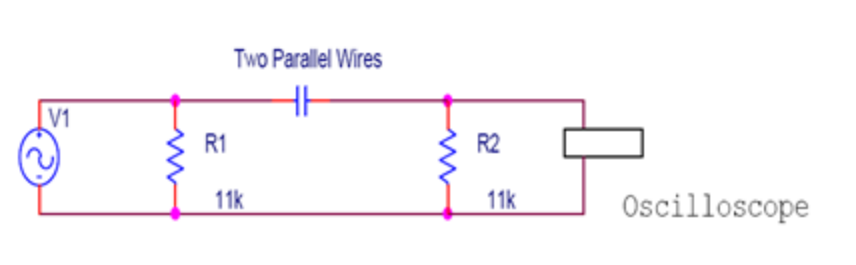
$$C = Q/V = \frac{2\pi l a \rho_s}{\frac{2\rho_s a}{\varepsilon_0}\ln\left(\left|\frac{a-D}{a}\right|\right)} = \boxed{\frac{\pi\varepsilon_0 l}{\ln\left(\left|\frac{a-D}{a}\right|\right)}}$$

Capacitance

- Estimate the capacitance of the setup below, assuming the wires have length $l = 10cm$ and $d = 3a$



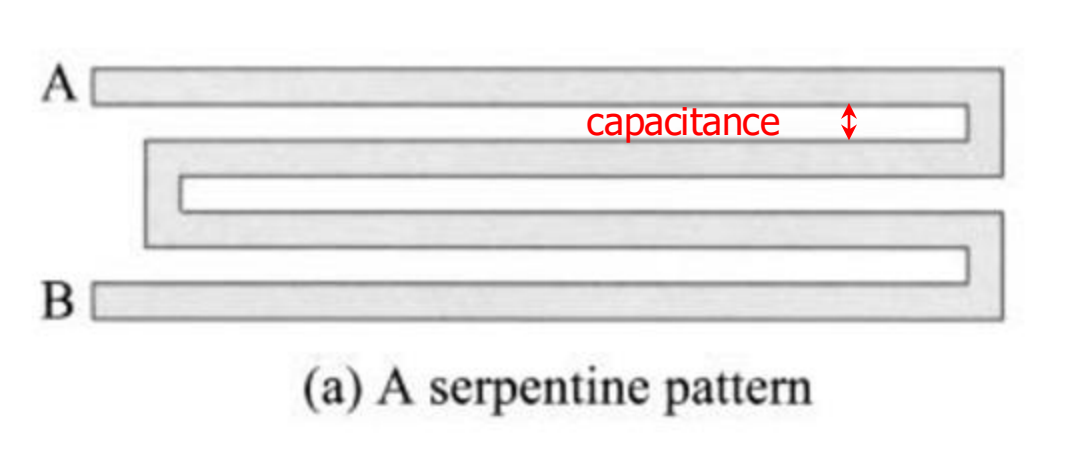




$$C = \frac{\pi \varepsilon_0 l}{\ln\left(\frac{d-a}{a}\right)} \approx 8\ \text{pF}$$

Capacitance

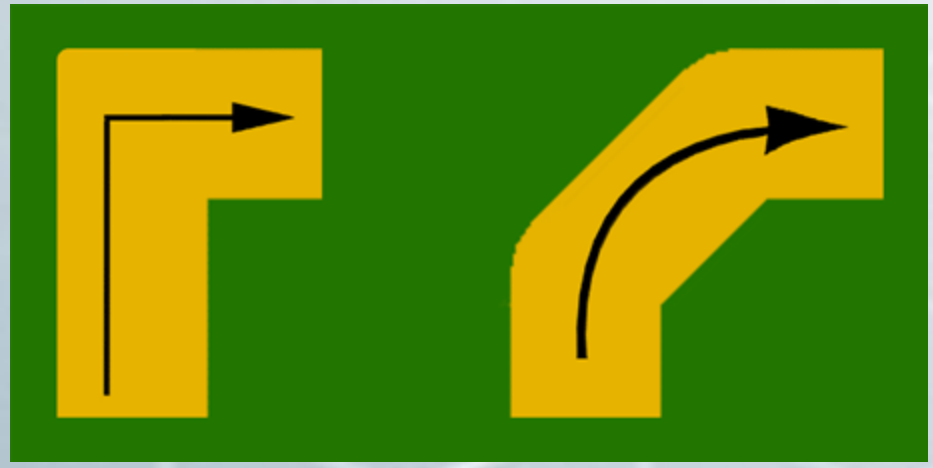
- In Ulaby, capacitance is generally described between two isolated conductors separated by a dielectric. But it also occurs between different parts of a single conductor.
- In circuit design, this unwanted effect is called “parasitic capacitance.”
- Circuit designers choose wire geometries to reduce this effect (and other effects)



(a) A serpentine pattern

Capacitance

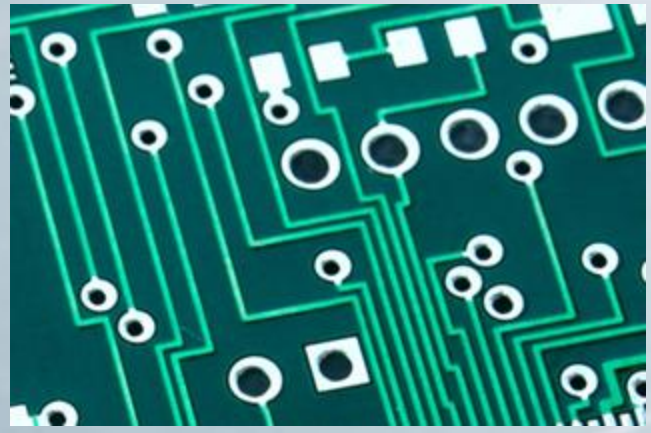
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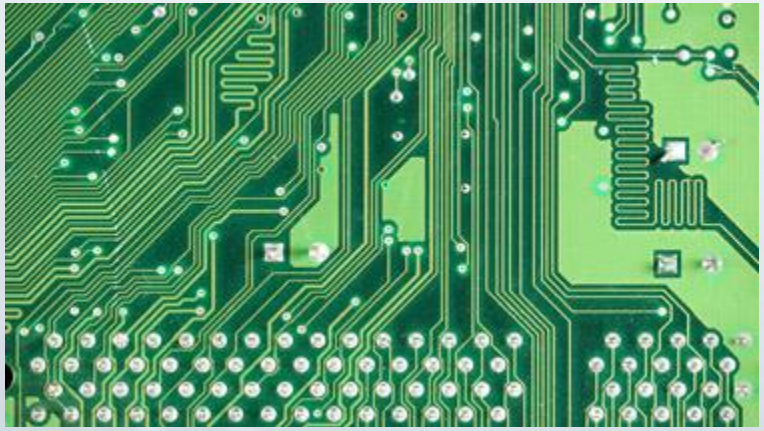


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Capacitance

- This has inspired changes in circuit design, such as the 45-degree angles in the PCB at right.
- This effect can be relevant for precise timing / digital circuit applications but is often overestimated





Hackaday

Wrap-Up

- HW6 out yesterday